AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

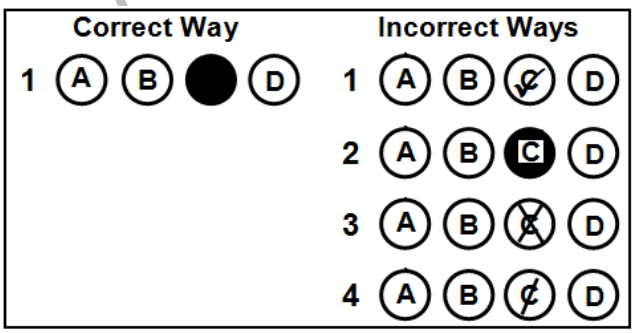
Computer Science Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only. From question number 13 to 30, there are two sections. Attempt EITHER section I (C Language) OR section II (Visual Basic).
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. The database object which holds data is called a

   A. table.
   B. form.
   C. query.
   D. report.

2. Consider the given database table.

StudentID	FirstName	LastName	Class
S200	Umar	Khan	9
S201	Ayesha	Mughal	10

   The number of records in the given database table is

   A. 1
   B. 2
   C. 3
   D. 4

3. Consider the given database table.

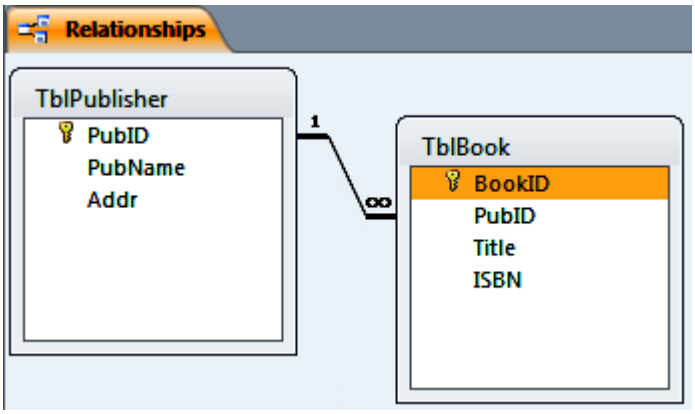
StudentID	FirstName	LastName	Class
S200	Umar	Khan	9
S201	Ayesha	Mughal	10
S202	Umar	Usman	10
S203	Affan	Bajwa	9
S204	Shanzay	Khan	10

   Which of the following fields should be selected as a primary key?

   A. Class
   B. LastName
   C. FirstName
   D. StudentID

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4. Consider the figure given below.



The relationship shown between the tables is

A. many-to-one relationship.
B. one-to-many relationship.
C. many-to-many relation.
D. one-to-one relationship.

5. A relation between two tables is created if both tables have at least one common

A. row.
B. field.
C. record.
D. primary key.

6. All of the following are field sizes for number data type in MS Access EXCEPT

A. byte.
B. float.
C. single.
D. integer.

7. Which of the following data types can store the longest text?

A. Hyperlink
B. Date/ Time
C. Memo
D. Text

8. In MS Access, the icon  is used for

A. descending sort.
B. ascending sort.
C. spell check.
D. selection.

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9. In MS Access, the keyboard shortcut to find the next match without using **Find and Replace** dialog box is

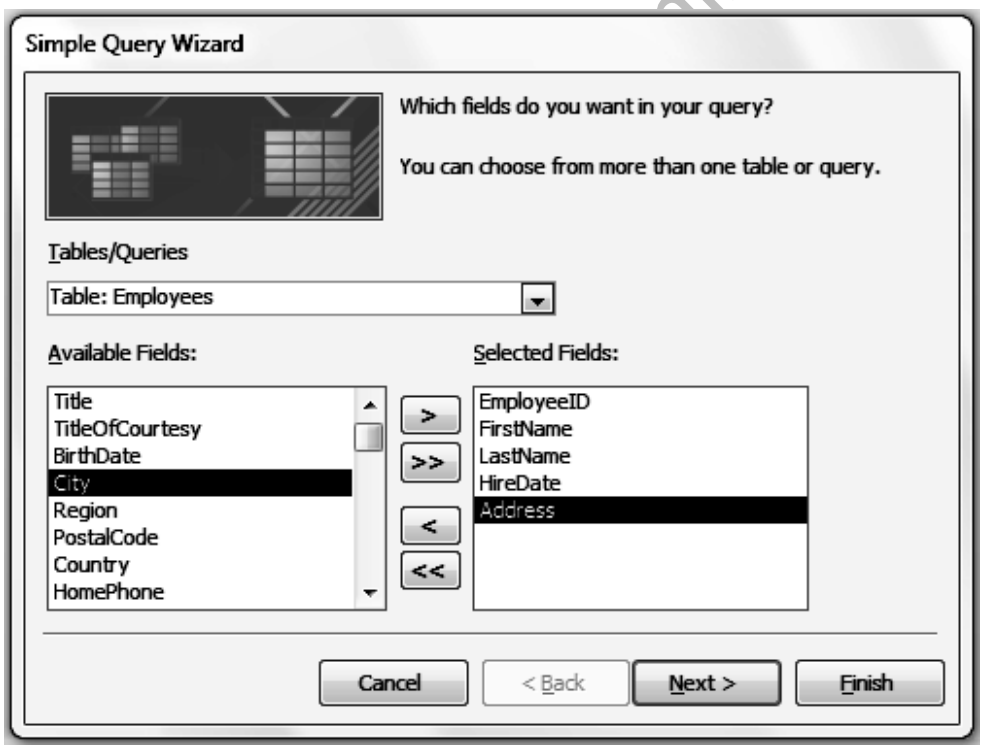
   A. Ctrl + F
   B. Shift + F
   C. Ctrl + F4
   D. Shift + F4

10. In which of the following MS Access form(s), each field appears on a separate line with a label to its left side?

   I. Tabular Form
   II. Columnar Form
   III. Datasheet Form

   A. I only
   B. II only
   C. I and II
   D. II and III

11. Consider the given simple query wizard.



If any field is double-clicked from **Selected Fields** then the

   A. field will be deleted from Selected Fields.
   B. next step in Simple Query Wizard will be shown.
   C. data of the field will be shown in a separate window.
   D. field will be moved from Selected Fields to Available Fields.

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12. In MS Access, a report is produced by combining data using

A. forms and queries.
B. tables and queries.
C. forms and creating relationships.
D. tables and creating relationships.

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ATTEMPT EITHER SECTION I OR II

Section I C Language

13. Which of the following is the file extension of C program?

A. .C
B. .EXE
C. .LIB
D. .OBJ

14. Which of the followings is TRUE for C language?

A. It is a highly structured language.
B. It is difficult for humans to understand.
C. It uses assembler for translation purposes.
D. It is a type of low level programming language.

15. Each statement in a C program is terminated by using a/ an

A. semicolon.
B. hash symbol.
C. forward slash.
D. inverted comma.

16. Read the given C program.

```
#include<stdio.h>
int main(void)
{
printf("Hello\nWorld");
return 0;
}
```

The output of the given C program will be

A	B	C	D
"Hello\nWorld"	Hello\nWorld	Hello World	Hello /nWorld

17. int main(void)

In the given command, int before the main function indicates that the main function will

A. return any float value.
B. return an integer value.
C. need any float argument.
D. need an integer argument.

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18. Which of the following C data types has the typical size of 64 bits?

A. Double
B. Float
C. Char
D. Int

19. The output of the given C language code will be

```
int    a = 25;
int    b = 2;
printf ("Result  =  %d" , 6 + a / 5 * b)
```

A. Result = 16
B. Result = 8.5
C. Result = 6 + a / 5 * b
D. Result = %6 + a / 5 * b

20. Which of the following is a logical operator in C language?

A. !
B. !=
C. = =
D. =

21. The output of printf (" %d ", 19 % 4) will be

A. 0.75
B. 3
C. 4
D. 4.75

22. Read the given C language code.

```
int h = 1;
while (h <= 5)
{
   printf("%d\n", h);
   h = h + 1;
 }
```

The output of the given C program will be

A	B	C	D
1 2 3 4	4 3 2 1	1 2 3 4 5	5 4 3 2 1

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23. The number of repetitions due to the given For loop statement are

for (a = 0; a <= 13; a = a + 2)

A. 6
B. 7
C. 13
D. 14

24. If the value of x is 3, then according to the given code, value of y will be

```
if (x < 3)
y = 5;
else if (x ==3)
y = 10;
else if (x > 3)
y = 15;
else
y = 20;
```

A. 5
B. 10
C. 15
D. 20

25. Study the given C language code.

```
char  Grade = 'b';
switch( Grade )

   {
      case 'A' : printf( "Excellent" );
      break;
      case 'B' : printf( "Very Good" );
      break;
      case 'C' : printf( "Good" );
      break;
      default  : printf( "Enter a Valid Grade" );
      break;
    }
return 0;
```

The output of the given code will be

A. Excellent
B. Very Good
C. Good
D. Enter a Valid Grade

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26. Which of the following statements is TRUE about the Do While loop?

A. It is a type of built-in functions in C language.
B. It is a type of selection statements in C language.
C. It tests condition after executing all statements inside the loop.
D. It tests condition before executing any statement inside the loop.

27. The C language instruction to call a user defined function named printMessage() within main() function is

A. printMessage()
B. printMessage() **;**
C. void printMessage(void)
D. void printMessage(void) **;**

28. If a user has defined a function named square() in C language that takes one integer argument, then the CORRECT way to declare this function is

A. int square (void) **;**
B. void square (int a)
C. integer square (void) **;**
D. void square (integer a)

29. Any function in C language by default returns a value of the type

A. double.
B. float.
C. char.
D. int.

30. Which of the following header files provides the prototypes for the file Input/Output (I/O) functions?

A. sting.h
B. stdio.h
C. conio.h
D. stdlib.h

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ATTEMPT EITHER SECTION I OR II

Section II Visual Basic

13. In Visual Basic, the default toolbar that appears below the menu bar is

A. standard.
B. debug.
C. form editor.
D. edit.

14. The keyword used to comment a code line in Visual Basic is

A. Rem.
B. Com.
C. Remark.
D. Comment.

15. Which of the following file extensions is used for ActiveX control files?

A. .OCX
B. .EXE
C. .FRM
D. .BAS

16. Which of the following supports an activated event?

A. Label
B. Button
C. Form
D. Text Box

17. Read the given code.

```
Dim x As Integer = 20
Dim y As Double = 3
Dim z As Double
z = x Mod y
```

The result of this code will be

A. 6
B. 2
C. 17
D. 23

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18. While considering the purpose of data types in Visual Basic, the appropriate data type to store the address of a person is

A. string.
B. long.
C. byte.
D. single.

19. Which of the following will give the result as 1?

A. Print 1 XOR 0
B. Print 1 And 0
C. Print Not 1And Not 0
D. Print Not 1 XOR Not 0

20. The number of byte(s) occupied by the integer data type is

A. 2
B. 1
C. 4
D. 8

21. While considering the purpose of data types in Visual Basic, the appropriate data type to store the alphabet H is

A. long.
B. byte.
C. string.
D. single.

22. The output of the given Visual Basic code will be

```
Dim strMyString, strMain
strMain = "TURKEY"
strMyString = Left(strMain, 3)
```

A. str
B. KEY
C. TUR
D. Main

23. A user has made a function named as Addnum() to add two numbers and return integer value. Which of the following is the CORRECT code line to call this function in Visual Basic program?

A. Public Function Addnum(num1, num2 As Integer) As Integer
B. Addnum(num1, num2)
C. Addnum(num1, num2 As Integer)
D. Public Function Addnum(num1, num2 As Integer)

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24. The conditional operator is used in IF…THEN…ELSE statement to return

A. the lowest of two values.
B. one of the two values depending on a certain condition.
C. one of the two values alternately.
D. the highest of two values.

25. Read the given Visual Basic code.

```
Dim count As Integer = 0
Do While count <= 10
count + = 1
Loop
```

After the loop ends, the value of 'count' will be

A. 0
B. 1
C. 10
D. 11

26. Read the given code.

```
If shift_code = "1" Then
shift_rate = hourly_rate * 1.1
ElseIf shift_code = "2" Then
shift_rate = hourly_rate * 1.2
ElseIf shift_code = "3" Then
shift_rate = hourly_rate * 1.3
Else
Print "Shift code error"
End If
```

If the hourly rate is 10 and shift_code is 3, then the output of the given code will be

A. 11
B. 12
C. 13
D. Shift code error

27. Which of the following codes should be used in click events for a button to go on next record placed within a form named datAuthors?

A. Recordset.MoveNext.datAuthors
B. datAuthors.Recordset.MoveNext
C. datAuthors.Recordset.Next
D. Recordset.Next.datAuthors

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28. Which of the following is used to keep track of the current record in an open database's table?

    A. Recordset
    B. Record pointer
    C. Snapshot
    D. Dynaset

29. All of the following are objects of ActiveX Data Object (ADO) model EXCEPT

    A. connection.
    B. command.
    C. field.
    D. table.

30. Which of the following is a VALID Visual Basic statement to draw a circle?

    A. Circle 150, 200, 75
    B. Circle (150, 200), 75
    C. Circle (150, 200)
    D. Circle 150, 200

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END OF PAPER

Please use this page for rough work

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